

Premise

The Engström menus are controlled by using the ComWheel. The ComWheel is the blue knob situated in the lower right-hand section of the screen. Turning the knob clockwise or counterclockwise will scroll through the menu items or change the settings. When a menu item or setting is highlighted, pushing the ComWheel will confirm the selected item.

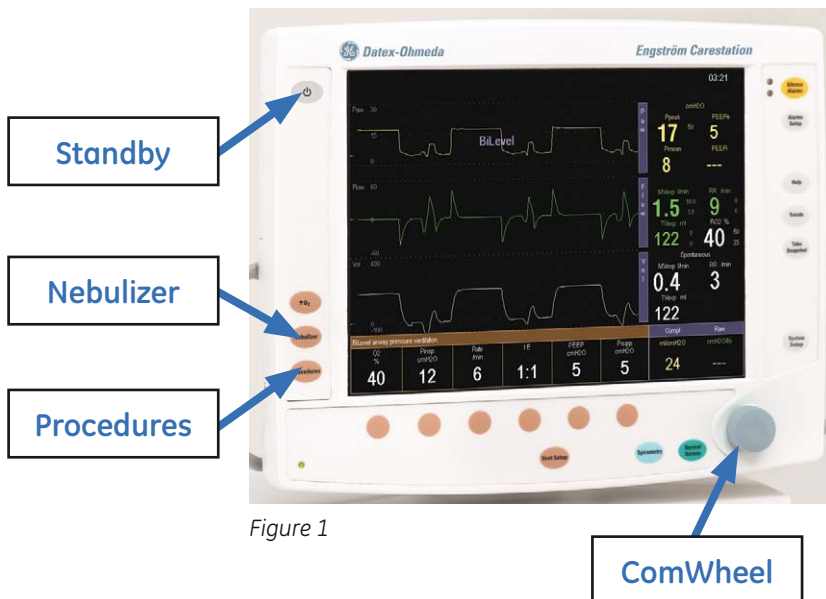


Figure 1

Standby

Monitoring and ventilation cease when the ventilator is placed into Standby. Follow the method for “Starting ventilation” to exit Standby.

WARNING The patient will not be ventilated when in Standby.

To enter Standby:

1. Disconnect the patient from the circuit.
2. Push the **Standby** button located in the upper left hand section of the display (Figure 1).
3. Select **Standby** (see the picture on the following page)

Note: If you select **Standby** without disconnecting the patient or if you occlude the circuit, the alarm “Patient connected” will appear on the screen.



Park circuit (only from software version 5.x)

Use this function to allow the patient circuit to be occluded without the ventilator alarming while in standby. This function allows the patient circuit to be hygienically protected while waiting to connect the patient. Removing the circuit occlusion will clear the Park Circuit status.

The message "Circuit Parked" appears on the screen while in this mode.

WARNING The patient will not be ventilated when the circuit is parked.

To enter Park Circuit:

1. Following the procedure to entering "Standby".
2. Occlude the patient circuit.
3. Select **Park Circuit** (see the picture above): the ventilator stops alarming.

As soon as the circuit is being opened, the ventilator turns to Standby position automatically and the alarms can be activated again.

Turning the system off

The Engström may only be turned off when in **Standby**. Follow the procedure for entering "**Standby**," and turn the system switch off.

Starting ventilation

To start ventilation:

1. Push the **Standby** button located on the upper left hand side of the display (Figure 1).
2. Select **Start Ventilation**.
3. Connect the circuit to the patient.

Automatic Checkout

To perform the pre-use Checkout, the ventilator must be in Standby (see Standby section) and the patient disconnected from the ventilator.

Checkout appears in **Patient Setup** menu.

1. Go to **Patient Setup** menu:

- If the ventilator is OFF:

Power the ventilator ON by moving the System Switch up. Select the patient type (Adult, Pediatric or Neonatal) (Figure 2), the **Patient Setup** menu will appear on the normal screen (Figure 3).



Figure 2



Figure 3

- If the ventilator is ON and in Standby:
Push the **System Setup** button on the right side of the display and select **Patient Setup**. The **Patient Setup** menu will appear on the screen.
2. Select **Checkout** (Figure 3).
 3. Attach the patient circuit to the ventilator and occlude the patient wye as shown in Figure 4.



Figure 4

4. Select **Start Check** (Figure 5).
 - The results appear next to each step as they are completed. 'Checkout complete' will appear and the highlight will move to **Delete Trends** when the checkout is done (Figures 5 and 6).

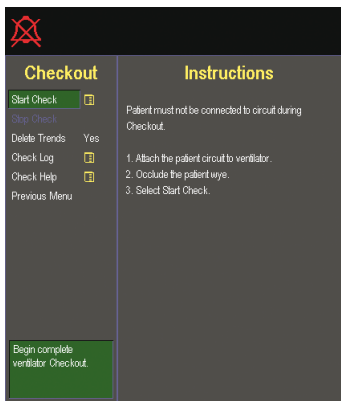


Figure 5

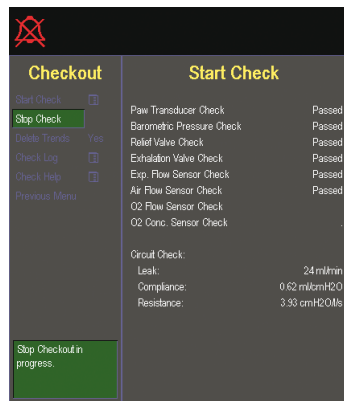


Figure 6

5. Select *Yes* to erase trends or *No* to retain the saved trends.
6. If one or more checks failed, select *Check Help* for troubleshooting tips.
7. Select *Previous Menu* once the checkout is successfully completed to return to the Patient Setup menu.

Important: If the Relief Valve Check fails during Checkout the ventilator cannot be used to ventilate a patient.

Suction routine

The Suction Procedure provides:

- Pre-oxygenation with 100% O₂ to the patient for 2 minutes
- Automatically switching to Standby for 2 minutes
- Post-oxygenation with 100% O₂ for 2 minutes.

Note: The suction procedure is not to be used with in-line suction devices. The suction procedure requires the patient to be disconnected in order for the procedure to operate correctly.

Suction Procedure:

1. The Procedure button is located on the left side of the display (Figure 1). Push the **Procedures** button.
2. Turn the ComWheel to select **Suction** (Figure 7).



Figure 7

3. The system will immediately start delivering 100% O₂ for 2 minutes or until the patient is disconnected from the ventilator. A message appears on the right top portion of the screen describing which step of the suction procedure is active (Figure 8).

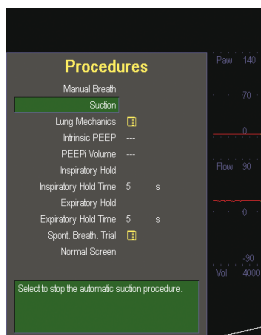


Figure 8

4. Disconnect the patient from the ventilator during the 2 minute pre-oxygenation stage.

A medium priority alarm will sound once after disconnecting the patient with no message. The system immediately enters Standby for 2 minutes or until the patient is reconnected to the ventilator (Figure 9). Suction the patient.

Note: If the patient is not disconnected during the first 100% O₂ phase, the suction procedure will cancel.



Figure 9

5. Reconnect the patient to the ventilator. The ventilator will cancel out of Standby and resume ventilating the patient using the setting established before the procedure. The ventilator will post-oxygenate using 100% O₂ for 2 minutes.

Note: To stop an active suction procedure during the 100% O₂ delivery, push the **Procedures** button and select **Suction** or push the O₂ button.

Expiratory valve maintenance

1. Put the ventilator in “Standby” or turn it off.
2. Remove the exhalation valve assembly from the ventilator by pressing down on the latch (Figure 10) and pulling the assembly away from the ventilator.



Figure 10

3. Remove the expiratory flow sensor from the exhalation valve assembly and set aside or clean.

4. Unscrew the water trap and empty.
5. Remove the diaphragm from the housing.
6. Sterilize the exhalation valve by autoclaving at 134° and cool to room temperature or by using one of the recommended procedures listed below.

WARNING: The following cleaning/disinfection agents have been validated for material compatibility only. The effectiveness of sterilization with these agents has not been validated.

Component	Cleaning / Disinfection agents						
	Ethyl Alcohol	Sporax II	Cidex Plus	NU-CIDEX	CIDEX OPA	Autoclave at 134°C	Mild detergent and warm water
Expiratory flow sensor	yes	yes	yes	no	yes	yes	yes
Exhalation valve housing (includes diaphragm, o-ring and water trap)	yes	yes	yes	no	yes	yes	yes

7. Reassemble the water trap and diaphragm on the exhalation valve housing.
8. Reassemble the dry exhalation valve housing and expiratory flow sensor.
9. Install the exhalation valve assembly onto the ventilator (see the picture below).
10. Complete a system Checkout prior to use on a patient.

Aeroneb PRO nebulizer

Note: Gas sampling and monitoring is suspended while the nebulizer is in use.

Assembling the nebulizer

1. Connect the nebulizer to the T-adapter by pushing the nebulizer firmly onto the adapter (Figure 11).



Figure 11

2. Connect the nebulizer and T-adapter into the inspiratory limb of the breathing circuit before the patient wye (Figure 12).



Figure 12

WARNING: Always maintain the nebulizer in a vertical orientation while in the patient circuit. This orientation helps prevent patient secretions and condensate from contaminating the aerosol generator of the nebulizer and ensures proper nebulization.

3. Attach the nebulizer cable connector to the nebulizer connection on the ventilator as shown in Figure 13, matching the red dots.



Figure 13

Filling the nebulizer

1. Open the filler cap tab on the nebulizer (Figure 14).



Figure 14

2. Use a pre-filled nebule or syringe to inject the medication into the filler port.
3. Close the filler cap tab.

Nebulization procedure

On the lower left hand side of the display, push the “**Nebulizer**” button (Figure 15).

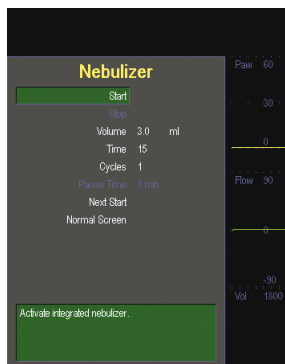


Figure 15

1. Using the ComWheel, select **Volume** or **Time** and change to the desired value. If **Volume** or **Time** is being selected, automatically **Time** or **Volume** will be set accordingly.
2. To deliver multiple nebulizer cycles, set the number of cycles and the **Pause Time** between cycles.
3. Select **Start**. A nebulization message appears in the upper right part of the screen (see picture).
4. To stop the nebulizer before the selected nebulization time has expired, push the **Nebulizer** button and select and confirm – **Stop**.

Disassembling the nebulizer

The nebulizer and T-adapter may remain in the patient circuit when not in use. The nebulizer may be removed from the T-adapter and replaced with a plug to avoid leaks.

1. To remove the connector, grasp it close to the ventilator and pull straight out.
2. Remove the nebulizer and T-adapter from the inspiratory limb of the patient breathing circuit. Reconnect the circuit.
3. Clean and sterilize the nebulizer and T-adapter as described in “*Aeroneb PRO nebulizer maintenance.*”

Aeroneb PRO nebulizer maintenance

Sterilize the nebulizer prior to the first use on a patient. Clean and sterilize the nebulizer between patients.

1. Remove the nebulizer and the adapters from the ventilator circuit.
2. Disassemble the nebulizer unit and adapters into individual components (Figure 16).



Figure 16

CAUTION: Do not autoclave or submerge the nebulizer cable.

3. Remove the filler cap from the nebulizer and empty any excess liquid.
4. Clean all of the parts in warm water and a mild liquid detergent.
5. Shake the excess water from the parts, and allow the parts to air dry.

CAUTION: Do not reassemble parts prior to autoclaving.

6. Sterilize the components.
 - To steam sterilize, autoclave the wrapped parts using steam sterilization pre-vacuum cycle at 132°C to 135°C for 3 to 4 minutes with a drying cycle.
 - To sterilize with hydrogen peroxide gas plasma, place wrapped parts in a STERRAD 100S System and use the long cycle.
7. Clean nebulizer by using one of the recommended procedures:

Component		Cleaning / Disinfection agents					
	Ethyl Alcohol	Sporax II	Cidex Plus	NU-CIDEX	CIDEX OPA	Autoclave at 134°C	Mild detergent and warm water
Aeroneb Pro Nebulizer	no	no	yes	yes	yes	yes	yes

CAUTION: Refer to the product labeling for STERRAD 100S System for specific instructions regarding proper operation. When the STERRAD method is used, high-level disinfection with CIDEX, NU-CIDEX or CIDEX OPA should not be carried out.